HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **MES recipes** adhering to corporate design standards for a biopharmaceutical client, ensuring alignment with high product quality and lean manufacturing goals.
- Performed **MES recipe authoring activities**, translating complex process requirements into functional MES content to support facility objectives.
- Participated in the **testing and computer system validation (CSV)** of MES recipes and associated configurations, ensuring compliance and system integrity.
- Contributed to the creation of key **MES deliverables**, including design documents and validation documentation, in collaboration with cross-functional teams.
- Managed **MES system administration** for third-party interfaces, including **SAP** and **LIMS**, facilitating seamless data exchange and operational continuity.
- Configured and built **MES system components** within Syncade, focusing on Equipment, Material Management, and Reporting functionalities to enhance manufacturing visibility.
- Automated data extraction and reporting tasks using **Python scripting**, processing approximately **1,500** daily transaction records for operational analysis.

SKILLS

MES Engineering – MES System Design, MES Recipe Authoring, Computer System Validation (CSV), Master Recipes, Process Controls, Information Systems

Programming & Scripting – Python, Visual Basic, SQL

Databases – Microsoft SQL Server, PostgreSQL, MySQL, Oracle

Core Technologies – Java, Spring Boot, REST APIs, AWS, Docker, Linux, Git, Jenkins, Jira, Confluence

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, pandas, NumPy, scikit-learn

Manufacturing Systems – SAP, LIMS, DeltaV, ERP Integration Support

PROJECTS

MES Recipe Development and Validation for Pharmaceutical Production

- Designed and implemented **MES master recipes** for a critical biopharmaceutical production line, focusing on process control and data integrity.
- Executed comprehensive **testing and validation protocols** for new and updated MES recipes, documenting results to meet regulatory compliance standards.
- Collaborated with process engineers and quality assurance teams to gather requirements and ensure MES content accurately reflected manufacturing workflows.

Automated Data Extraction and Reporting Pipeline

- Developed a **Python-based ETL pipeline** to extract, transform, and load data from enterprise systems into a centralized data warehouse, processing ~**800** GB of data weekly.
- Created automated SQL scripts for data validation and reconciliation across multiple databases (**PostgreSQL**, **MySQL**), identifying and resolving data discrepancies.
- Built scheduled reports using **Power BI** to visualize key operational metrics for business stakeholders, improving decision-making speed by an estimated **15%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)